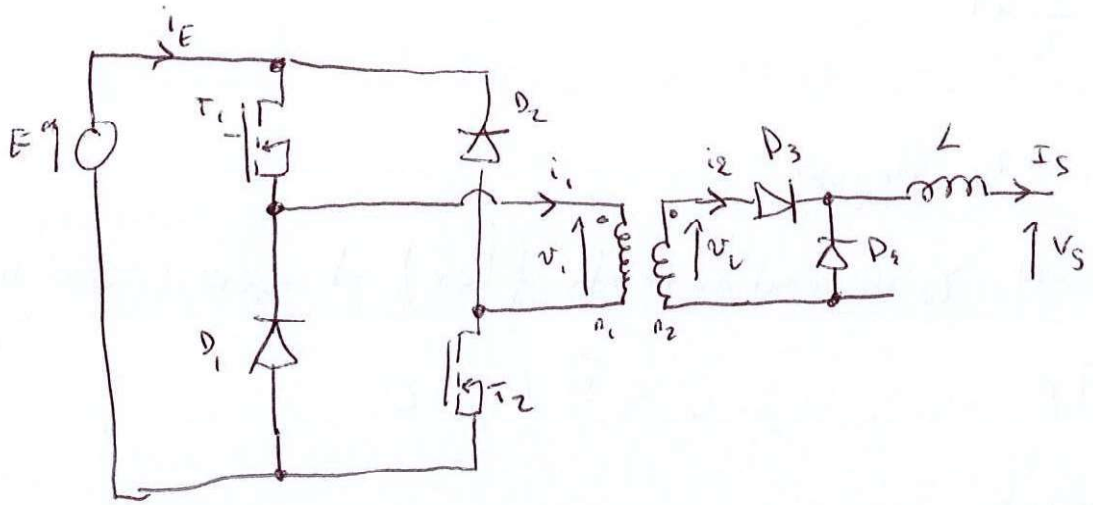
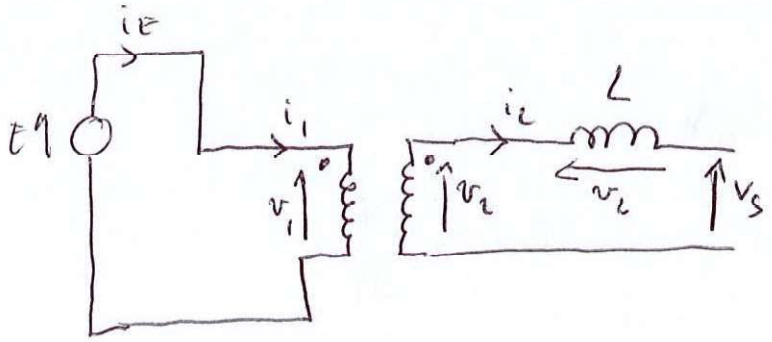
Direct Forward Converter (Asymmetric FORWARD)



1.1) (i) $v_1 = n_1 \dfrac{d\varphi}{dt}$ $\qquad v_2 = n_2 \dfrac{d\varphi}{dt}$

(ii) $n_1 i_1 - n_2 i_2 = \mathcal{R}\varphi$

1.2) $0 < t < \alpha T$: $\varphi(t=0) = 0$ $\qquad T_1$ et T_2 closed



(i) $v_1 = E$

(ii) $v_2 = \dfrac{n_2}{n_1} v_1 = \dfrac{n_2}{n_1} E > 0 \Rightarrow \begin{cases} D_3 & \text{forward} \\ D_4 & \text{reverse} \end{cases}$

(iii) $v_L = v_2 - V_S = \dfrac{n_2}{n_1} E - V_S$

(iv) $v_1 = E = n_1 \dfrac{d\varphi}{dt}$ or $\varphi(t=0) = 0 \Rightarrow \varphi(t) = \dfrac{E}{n_1} t$

(v) $i_2 = i_L$.

(vi) $n_1 i_1 - n_2 i_2 = \mathcal{R}\varphi \Rightarrow i_1 = \frac{n_2}{n_1} i_2 + \frac{\mathcal{R}}{n_1}\varphi$

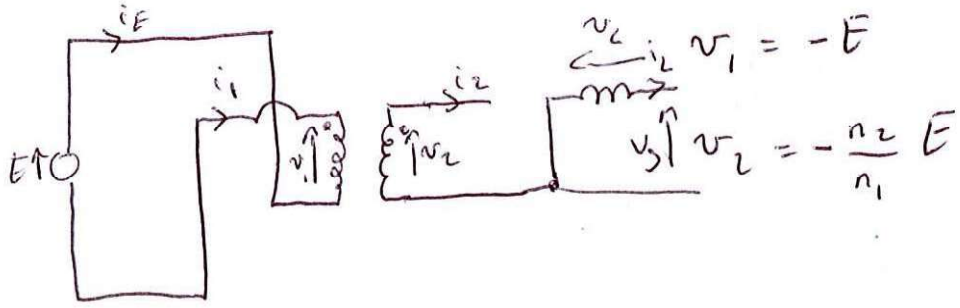
(vii) $\varphi_0 = \varphi(t=\alpha T) = \frac{E}{n_1}\alpha T.$

1.3) $\alpha T < t < T$ T_1 et T_2 open

(i) à $t=\alpha T$ $\varphi(t=\alpha T) = \varphi_0 > 0$ continuity of flux $\Rightarrow$ et $i_2 \geq 0$ (current in D_3)

$n_1 i_1 - n_2 i_2 = \mathcal{R}\varphi \Rightarrow i_1 = \frac{n_2}{n_1} i_2 + \mathcal{R}\varphi > 0$

$\Rightarrow D_1$ et D_2 forward



$v_1 = -E$

$v_2 = -\frac{n_2}{n_1}E$

(continuity of current in L)

$v_2 < 0 \Rightarrow \begin{cases} D_3 & \text{reverse} \\ D_4 & \text{forward} \end{cases}$

(ii) $v_L = -V_3 < 0 \rightarrow i_L \searrow$

(iii) $\varphi(t=\alpha T) = \varphi_0$ et $v_1 = -E = n_1 \frac{d\varphi}{dt}$

$\Rightarrow \varphi(t) = -\frac{E}{n_1}(t-\alpha T) + \varphi_0$

(iv) $\varphi(t_0) = 0 \Rightarrow -\frac{E}{n_1}(t_0 - \alpha T) + \varphi_0 = 0$

$\Rightarrow t_0 = \alpha T + \frac{\varphi_0 n_1}{E} = \alpha T + \alpha T = 2\alpha T$

$t_0 < T \Rightarrow 2\alpha T < T \Rightarrow \boxed{\alpha < \frac{1}{2}}$

(✓) Pour $t > t_0$:

* if D_1 and D_2 forward :

$v_1 = -E < 0 \Rightarrow \varphi \downarrow \Rightarrow \varphi < 0$

$n_2 i_2 = n_1 i_1 - \Re \varphi > 0 \rightarrow D_3$ forward $(i_1 \geq 0)$

or $v_2 = -\frac{n_2}{n_1} E < 0 \Rightarrow D_4$ forward $\Rightarrow v_2 = 0$ absurde.

$\Rightarrow$ $\boxed{D_1 \text{ et } D_2 \text{ reverse}}$.

$\hookrightarrow i_1 = 0$

* Si $\varphi \downarrow \Rightarrow \varphi < 0$ (continuity of flux at $t = t_0$)

$n_2 i_2 = n_1 i_1 - \Re \varphi > 0$ ($i_1 \geq 0$ avec D_1) $\rightarrow D_3$ forward.

$v_2 = n_2 \frac{d\varphi}{dt} < 0 \rightarrow D_4$ forward $\rightarrow v_2 = 0 \rightarrow$ pb !!!

* Si $\varphi \uparrow \Rightarrow \varphi > 0$

$n_1 i_1 = \Re \varphi + n_2 i_2 > 0$ ($i_2 \geq 0$ avec D_3)

donc D_1 conduit $\Rightarrow D_2$ forward $\Rightarrow v_1 = -E = n_1 \frac{d\varphi}{dt} < 0$

$\rightarrow \varphi \downarrow$ pb !!!

$\Rightarrow$ $\boxed{\varphi = \text{constant} = 0}$.

$\rightarrow \frac{d\varphi}{dt} = \boxed{0 = v_1 = v_2}$

$\Rightarrow$ $\boxed{D_3 \text{ et } D_4 \text{ forward}}$ $(i_3 > 0)$

$\boxed{i_1 = i_2 = 0}$

1.4) (i) $\langle v_L \rangle = 0 \Rightarrow \left(\frac{n_2}{n_1} E - V_S\right)\alpha - V_S(1-\alpha) = 0$

$$\Rightarrow \frac{n_2}{n_1}\alpha E = V_S$$

$$\Rightarrow \frac{n_2}{n_1} = \frac{V_S}{\alpha E} = 0{,}2.$$

(ii) $P_S = V_S \langle i_L \rangle \Rightarrow I_S = \frac{P_S}{V_S} = 25\,A.$

(iii) $\alpha T < t < T$: $v_L = -V_S = L\frac{di_L}{dt}$

$$\Rightarrow \Delta i_L = \frac{V_S(1-\alpha)T}{L} \Rightarrow L = \frac{V_S(1-\alpha)}{\Delta i_L f} = 0{,}23\,mH.$$

(iv) $I_{1_{max}} = i_1(t=\alpha T) = \frac{n_2}{n_1} i_{L_{max}} + \cancel{\frac{R\varphi}{n_1}}$ → négligé

$$= \frac{n_2}{n_1} \times 1{,}025\, I_S$$

$$= 5{,}1\,A.$$

2.1) $0 < t < \alpha T$: $\underset{E}{v_1} = L_1 \frac{di_m}{dt}$ à $t=0$, $i_m(t=0) = 0$ $(\varphi(t=0)=0)$

$$I_{m_{max}} = \frac{E}{L_1}\alpha T < I_{1_{max}}/20$$

$$\Rightarrow L_1 > \frac{20 E \alpha T}{I_{1_{max}}} = 9{,}4\,mH.$$

2.2) $0 < t < \alpha T$: $\begin{cases} i_1 = \frac{n_2}{n_1} I_S \\ i_2 = I_S \end{cases}$ (en négligeant i_m et Δi_L)

$\alpha T < t < T$: $\begin{cases} i_2 = 0 \\ i_1 = 0 \end{cases}$

$$I_{1,eff} = \frac{n_2}{n_1} I_s \sqrt{\alpha}$$

$$I_{2,eff} = I_s \sqrt{\alpha}$$

2.3) $$S_B = \frac{S_{Tw}}{k_B} = \frac{1}{k_b \delta}\left(n_1 I_{1,eff} + n_2 I_{2,eff} \right)$$

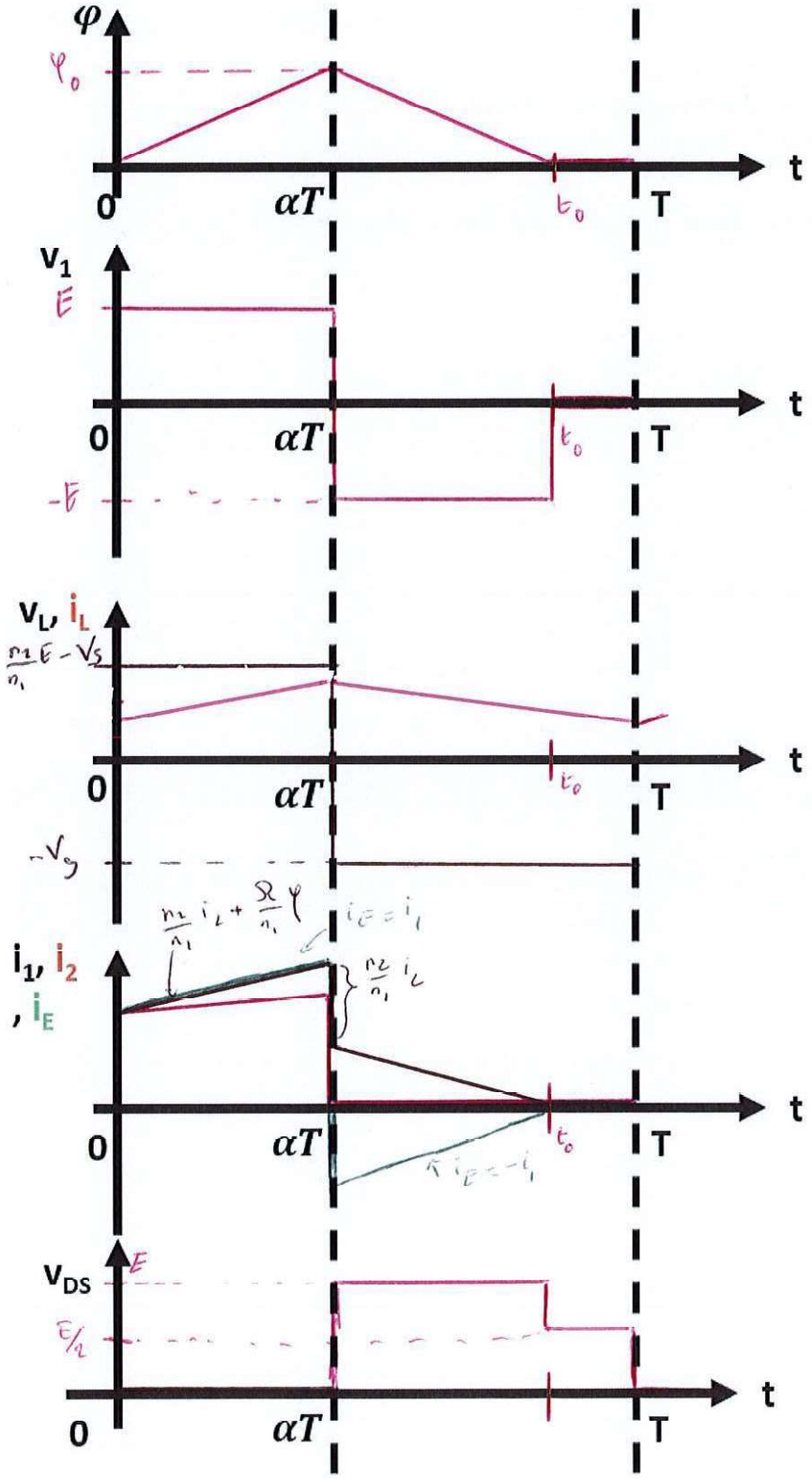
$$= \frac{1}{k_b \delta}\left(n_2 I_s \sqrt{\alpha} + n_2 I_s \sqrt{\alpha} \right)$$

$$= \frac{2 n_2 I_s \sqrt{\alpha}}{k_b \delta}$$

2.4) $$P_{max} = A_e B_{max} = \frac{E}{n_1} \alpha T$$

$$\Rightarrow A_{e_{min}} = \frac{E\alpha}{B_{max} n_1 f}$$

2.5) $$A_e S_B > \frac{2 n_2 I_s \overset{V_s}{E\alpha} \sqrt{\alpha}}{k_b \delta B_{max} n_1 f} = \frac{2 P_s \sqrt{\alpha}}{k_b \delta B_{max} f}$$


φ
φ0
0
αT
t0
T
t
V1
E
-E
VL, iL
n2/n1 E - Vs
-Vg
i1, i2
, iE
n2/n1 iL + R/n1 φ
iE = i1
n2/n1 iL
iE = -i1
VDS
E/2